

Comprehensive Review of Low-Power VLSI Design Techniques: Challenges, Cryptographic Implementations, and Future Prospects

1st Mr. Ayush D Bhuskat
Dept. of Electronics
NIELIT
Aurangabad, India
ayushbhuskat@gmail.com

2nd Mr. Aditya S Mandlik
Dept. of Electronics
NIELIT
Aurangabad, India
adityamandlik92@gmail.com

3rd Mr. Chaintanya N Kadadas
Dept. of Electronics
NIELIT
Aurangabad, India
chaitanya@nielit.gov.in

4th Mr. Lakshman Korra
Dept. of Computer Science
NIELIT
Aurangabad, India
lakshman@nielit.gov.in

5th Mr. Jayaraj U Kidav
Dept. of Electronics
NIELIT
Aurangabad, India
jayaraj@nielit.gov.in

6th Ms. Manjiri A Lavadkar
Dept. of Computer Science
NIELIT
Aurangabad, India
manjirilavadkar9@gmail.com

Abstract—The increasing demand for secure and energy-efficient computing has made low-power cryptographic hardware a crucial aspect of modern VLSI design. Achieving an optimal balance between power efficiency and security requires the implementation of effective power management techniques that minimize power dissipation while preserving system integrity. This paper presents an extensive review of various low-power VLSI design strategies, including dynamic voltage and frequency scaling (DVFS), clock gating, and power gating, and explores their role in cryptographic hardware implementations. Additionally, this study examines the trade-offs between power optimization and security, focusing on vulnerabilities such as power analysis attacks and presenting key mitigation techniques to counteract these threats. Furthermore, the paper discusses emerging technologies such as physically unclonable functions (PUFs) and AI-driven cryptographic power optimization as innovative solutions to enhance both power efficiency and security in VLSI circuits. By integrating low-power methodologies with security-centric optimizations, this research provides insights into the future development of next-generation VLSI systems, enabling efficient and secure hardware architectures that address the evolving challenges of power consumption and cryptographic security.

Index Terms—low power VLSI design , Power optimization, VLSI circuit design

I. INTRODUCTION

Minimizing power consumption in VLSI circuits is essential for optimizing energy efficiency, reducing operational costs, and improving the longevity of battery-powered devices. Various low-power design methodologies have been explored to achieve these objectives, incorporating circuit-level

and system-level optimizations. However, integrating cryptographic processes into these power-constrained systems adds complexity due to the computational intensity of encryption and decryption, as well as the security concerns associated with cryptographic implementations. Researchers have investigated power-efficient cryptographic mechanisms to address these challenges while ensuring high security standards.

The trade-off between power consumption and security is a significant issue in modern VLSI design. Cryptographic hardware is highly susceptible to power-based side-channel attacks, where an adversary can extract sensitive information by analyzing power variations during encryption and decryption operations. Several countermeasures, such as power randomization and secure clock gating, have been proposed to mitigate these threats while maintaining low power dissipation.

This paper provides a comprehensive review of low-power VLSI techniques and their application in cryptographic hardware. It discusses key power-saving methodologies, including dynamic voltage and frequency scaling (DVFS), clock gating, and power gating, while evaluating their impact on cryptographic security. The study also explores emerging trends, such as AI-driven cryptographic power optimization and physically unclonable functions (PUFs), which offer potential solutions for developing highly efficient and secure VLSI architectures. By integrating low-power methodologies with security-driven optimizations, this research aims to contribute to the advancement of next-generation VLSI systems that achieve an optimal balance between power efficiency and cryptographic security.

II. LITERATURE REVIEW

Low-power VLSI design has been a significant area of research due to the increasing demand for energy-efficient and high-performance semiconductor devices. Several studies have explored different techniques to reduce power consumption at various levels of VLSI design, from device-level optimizations to architectural strategies.

Pedram [1] discussed design technologies for low-power VLSI, highlighting the importance of power management techniques such as clock gating and dynamic voltage scaling (DVS). Rabaey et al. [2] provided a comprehensive analysis of low-power digital circuit design, emphasizing energy-efficient architectures and power-aware synthesis methodologies.

With the increasing demand for secure hardware, researchers have explored the integration of cryptographic techniques in low-power VLSI. Kocher et al. [3] introduced side-channel attack countermeasures, demonstrating the trade-offs between security and power consumption. More recent studies have focused on lightweight cryptography, where power-efficient encryption algorithms are designed for resource-constrained devices [4].

Emerging trends in hardware security and power optimization have led to the development of secure clock gating and power gating techniques for cryptographic hardware. These methods aim to minimize unnecessary power consumption while maintaining the integrity of cryptographic operations. Several researchers have proposed low-power secure multipliers and energy-efficient AES implementations to balance power efficiency and security [5].

Despite these advancements, power vs. security trade-offs remain a critical challenge. Aggressive power reduction techniques may inadvertently introduce vulnerabilities, making hardware susceptible to attacks. Therefore, future research must focus on developing power-aware security mechanisms that optimize both energy efficiency and resilience against threats.

III. POWER CONSUMPTION IN VLSI CIRCUITS

Power consumption in VLSI circuits is a crucial factor affecting system performance, energy efficiency, and overall reliability. In cryptographic hardware, power optimization is even more critical due to the intensive nature of encryption and decryption computations, which significantly impact energy consumption. Managing power dissipation effectively requires a thorough understanding of different power loss mechanisms and their influence on secure circuit designs [1]

- Dynamic Power Consumption
- Static Power Consumption
- Short-Circuit Power Consumption

A. Dynamic Power

Dynamic power consumption in VLSI circuits is primarily caused by the switching activity of transistors during circuit operation. It is given by the equation:

$$P_{\text{dynamic}} = \alpha C V^2 f$$

where α is the switching activity factor, C is the load capacitance, V is the supply voltage, and f is the operating frequency. Dynamic power can be reduced by lowering the supply voltage, optimizing clock gating, and minimizing unnecessary transitions in logic circuits.

Dynamic power consumption primarily results from transistor switching activities, which depend on parameters such as capacitance, supply voltage, and clock frequency. Cryptographic computations, particularly key generation and modular arithmetic operations, contribute to high switching activity, leading to elevated dynamic power usage. To mitigate this, techniques such as operand isolation, optimized substitution-box architectures, and dynamic voltage and frequency scaling (DVFS) are commonly applied [2]. These methods help control power consumption while ensuring the computational integrity of cryptographic operations.

B. Static Power

Static power consumption in VLSI circuits arises due to leakage currents that flow even when the circuit is in an idle state, making it a significant concern in nanometer-scale technologies. As transistor sizes shrink, leakage components such as subthreshold leakage, gate leakage, and junction leakage become more pronounced, leading to unnecessary power dissipation. This not only reduces energy efficiency but also impacts battery life and thermal stability, especially in low-power and portable devices. Effective techniques such as power gating, multi-threshold CMOS (MTCMOS), and body biasing are commonly used to minimize static power consumption.

C. Short-Circuit Power

Short-circuit power dissipation happens when the two PMOS and NMOS transistors also switch simultaneously at signal transitions to form a direct current path from supply to ground. This is usually due to slow input transition times. Short-circuit power can be minimized by minimizing input signal rise and fall time, the transistor being optimized Sizing and applying buffer insertion methods to enhance signal quality. integrity. While short-circuit power tends to be lower than dynamic and static power, it can nevertheless make a valuable contribution in high-speed circuits.

IV. LOW-POWER DESIGN TECHNIQUES

Low-power design techniques are essential in VLSI to minimize power dissipation while maintaining performance and reliability. These techniques are applied at different levels of design, including circuit, logic, architecture, and system levels. Circuit-level methods like clock gating and power gating help reduce dynamic and static power consumption, while logic-level optimizations focus on minimizing unnecessary computations. At the architecture and system levels, efficient memory management, asynchronous design, and adaptive power control further enhance power efficiency. the following are some key techniques used to minimize power consumption in VLSI circuits:

- Dynamic Voltage and Frequency Scaling (DVFS)
- Clock Gating
- Leakage Current Reduction

A. Dynamic Voltage and Frequency Scaling (DVFS)

Dynamic Voltage and Frequency Scaling (DVFS) is a power optimization technique that adjusts the supply voltage and operating frequency of a circuit based on workload requirements. Since dynamic power dissipation is proportional to the square of the supply voltage and directly proportional to frequency, reducing these parameters significantly lowers power consumption. However, lowering voltage can impact circuit performance, requiring a careful trade-off between power savings and processing speed. DVFS is widely used in modern processors, embedded systems, and mobile devices to optimize energy efficiency without compromising functionality. It is a key strategy in low-power VLSI design, helping to balance performance and power consumption dynamically.

B. Clock Gating

Clock Gating is a widely used technique to reduce dynamic power dissipation by disabling the clock signal for inactive circuit blocks. Since switching activity contributes significantly to power consumption, preventing unnecessary toggling of registers and logic elements helps minimize dynamic power. This technique is implemented using gating logic that selectively enables or disables the clock based on control signals. Clock gating is commonly used in processors, memory controllers, and other digital circuits to improve power efficiency while maintaining performance.

C. Power Gating

Power Gating is a technique used to reduce static power dissipation by completely turning off power to inactive circuit blocks. Unlike Clock Gating, which only disables the clock signal, Power Gating disconnects the supply voltage using power switches (sleep transistors). This helps minimize leakage power, which is a major concern in deep submicron technologies. Power Gating is widely implemented in modern processors, FPGAs, and low-power SoCs to improve energy efficiency without significantly impacting performance.

Power Gating Impact on Leakage Power:

$$P_{\text{leakage-PG}} = V_{DD} \times I_{\text{leakage-PG}} \quad (1)$$

where:

- $P_{\text{leakage-PG}}$ = Leakage power after applying power gating
- V_{DD} = Supply voltage
- $I_{\text{leakage-PG}}$ = Leakage current after power gating is applied

D. Leakage Current Reduction across Different Levels of Design

Leakage current is a major concern in low-power VLSI design, especially in advanced semiconductor technologies where static power dissipation significantly affects overall

power efficiency. As transistor sizes decrease, leakage power becomes more dominant, necessitating effective reduction techniques to ensure energy-efficient operation. Various strategies, such as high-threshold voltage transistors, power gating, and adaptive body biasing, are employed to control leakage without compromising performance. These techniques are applied across different levels of design, each contributing to minimizing leakage in specific ways.

These different levels of leakage reduction techniques are as follows:

- Device-Level Techniques
- Circuit-Level Techniques
- Architecture-Level Techniques
- System-Level Techniques

V. CRYPTOGRAPHIC HARDWARE POWER OPTIMIZATION TECHNIQUES

With the increasing demand for secure and energy-efficient VLSI designs, optimizing power consumption in cryptographic hardware has become a critical challenge. Cryptographic operations, such as encryption, decryption, and key management, involve intensive computations, leading to high power dissipation. Additionally, power variations in cryptographic circuits can expose them to side-channel attacks, compromising security. To address these challenges, several power optimization techniques have been developed to enhance the energy efficiency of cryptographic hardware while maintaining strong security. The following are some key techniques used for power-efficient cryptographic implementations:

- Lightweight Cryptography
- Energy-Aware Cryptographic Hardware
- Dynamic Power Randomization
- Secure Clock Gating

These techniques ensure that cryptographic hardware remains both energy-efficient and resistant to security vulnerabilities, making them essential for modern low-power VLSI design.

A. Lightweight Cryptography

Lightweight cryptography is designed for power-constrained devices such as IoT nodes, RFID tags, and embedded systems, where traditional cryptographic algorithms are too resource-intensive. These algorithms aim to reduce computational complexity, memory usage, and power consumption while maintaining a sufficient level of security. Techniques such as simplified key schedules, reduced-round encryption, and hardware-friendly operations are commonly used. Examples of lightweight cryptographic algorithms include PRESENT, SIMON, and SPECK, which are optimized for low-power VLSI implementations, ensuring security with minimal energy overhead.

B. Energy-Aware Cryptographic Hardware

Energy-aware cryptographic hardware focuses on optimizing encryption and decryption processes to reduce power consumption while maintaining high security. Techniques such

as hardware acceleration, optimized logic design, and reduced bit-width operations help in minimizing energy usage. Additionally, specialized low-power architectures for cryptographic algorithms like AES, RSA, and ECC ensure efficient execution with minimal power overhead. These optimizations are crucial for battery-operated and embedded systems where power efficiency is a priority.

C. Secure Clock Gating

Secure clock gating is an advanced power-saving technique that selectively disables the clock signal in cryptographic hardware without compromising security. While traditional clock gating reduces dynamic power by turning off inactive circuit sections, it can unintentionally leak information through side-channel attacks. Secure clock gating mitigates this risk by introducing random clocking variations, balancing switching activity, and ensuring uniform power consumption patterns. This approach effectively minimizes power dissipation while preventing attackers from exploiting power fluctuations to extract cryptographic keys, making it an essential technique for low-power and secure VLSI design.

VI. CHALLENGES IN LOW-POWER AND SECURE VLSI DESIGN

Despite significant advancements in low-power VLSI design, several challenges persist as technology scales down to deep submicron and nanometer levels. These challenges arise due to the increasing complexity of circuits, higher integration density, and the need for optimized power-performance trade-offs. Integrating low-power techniques with cryptographic security in VLSI design introduces several challenges. Integrating low-power techniques with cryptographic security in VLSI design introduces several challenges. Some of the key challenges are:

A. Power vs. Security Trade-off

In VLSI design, achieving both low power consumption and strong security is a challenging task, as these two goals often conflict. Many power-saving techniques, such as dynamic voltage scaling (DVS), clock gating, and power gating, reduce power dissipation but can also introduce side-channel vulnerabilities, making cryptographic hardware susceptible to attacks. On the other hand, security-enhancing methods like power randomization and masking techniques can increase power consumption, creating a trade-off between efficiency and protection. Balancing this trade-off requires careful optimization of power-aware security techniques, ensuring that cryptographic implementations remain both energy-efficient and resistant to attacks.

B. Leakage Power in Secure Circuits

Leakage power is a major concern in cryptographic hardware, especially in advanced VLSI technologies where transistor sizes are shrinking. Even when a circuit is idle, subthreshold leakage and gate leakage contribute to static power dissipation, which can be exploited in side-channel attacks. Cryptographic hardware, such as AES and RSA implementations,

must be designed to minimize leakage without compromising performance or security. Techniques like multi-threshold CMOS (MTCMOS), body biasing, and power gating help in reducing leakage power while maintaining data confidentiality. However, optimizing for both low leakage and strong security remains a critical design challenge.

C. Performance Overhead

Integration of power optimization methods into cryptographic hardware will have performance overhead, affecting encryption and decryption speed and efficiency techniques. Techniques such as dynamic voltage scaling (DVS) and power gating save power but could lead to lags behind in processing and higher latency, which can be problematic to real-time applications. In addition, security-improving methods such as masking and power randomization add additional calculations that weaken system performance. The problem is how to produce good cryptographic power-minimizing circuits that do not degrade substantially throughput and response time, and a balance between processing speed and energy efficiency.

D. Hardware Complexity

Integrating both low-power techniques and cryptographic security in VLSI circuits increases hardware complexity, impacting design scalability and manufacturing costs. Power-efficient cryptographic implementations require additional control logic, power management units, and security countermeasures, leading to increased transistor count and area overhead. Techniques such as secure clock gating, power randomization, and leakage reduction introduce design challenges, making circuit implementation more intricate. Furthermore, optimizing power without compromising security demands advanced design methodologies and verification processes, increasing development time and resource requirements. Balancing power, security, and hardware complexity remains a critical challenge in modern VLSI design.

E. Reliability Concerns

Power optimization techniques in cryptographic hardware can introduce reliability issues, affecting the correctness and stability of operations. Techniques like dynamic voltage scaling (DVS) and power gating can cause timing variations, signal integrity issues, and increased susceptibility to noise, potentially leading to errors in cryptographic computations. Additionally, aggressive power reduction strategies may impact device lifetime and aging effects, reducing long-term reliability. Ensuring secure and error-free cryptographic operations while maintaining low power consumption requires robust design strategies, error correction mechanisms, and careful power management to mitigate these risks.

F. Scalability for Emerging Technologies

As VLSI technology evolves, designing secure and power-efficient cryptographic hardware for emerging applications such as AI, IoT, 5G, and quantum computing presents new challenges. Traditional low-power techniques may not scale

effectively with next-generation semiconductor technologies, requiring innovative approaches to balance power, performance, and security. Additionally, heterogeneous computing architectures and advanced cryptographic algorithms demand efficient power management strategies to handle increasing workloads while maintaining security. Future research must focus on scalable low-power cryptographic designs that can adapt to the growing complexity of modern VLSI systems.

VII. EMERGING TECHNOLOGIES AND FUTURE TRENDS

As VLSI technology evolves, designing secure and power-efficient cryptographic hardware for emerging applications such as AI, IoT, 5G, and quantum computing presents new challenges. Traditional low-power techniques may not scale effectively with next-generation semiconductor technologies, requiring innovative approaches to balance power, performance, and security. Additionally, heterogeneous computing architectures and advanced cryptographic algorithms demand efficient power management strategies to handle increasing workloads while maintaining security. Future research must focus on scalable low-power cryptographic designs that can adapt to the growing complexity of modern VLSI systems. As power efficiency becomes a primary concern in VLSI design, several emerging technologies and future trends are shaping the development of next-generation low-power circuits. FinFET and GAAFET transistors are replacing traditional planar MOSFETs, offering reduced leakage and improved performance at lower power levels. 3D-ICs and heterogeneous integration allow for stacking multiple chips, reducing interconnect power losses and improving energy efficiency. Near-threshold and subthreshold computing are gaining traction by operating circuits at ultra-low voltages to achieve significant power savings. Additionally, AI-driven power management techniques leverage machine learning to optimize dynamic voltage scaling and power gating strategies in real-time. Other promising advancements include spintronics, quantum computing, and novel memory architectures such as memristors and RRAM, which offer alternative approaches for energy-efficient processing. These technologies collectively pave the way for highly power-efficient VLSI designs, addressing the challenges of modern semiconductor scaling. Following are some of the emerging technologies that contribute to low-power VLSI design:

VIII. SUMMARY

This paper provides a comprehensive review of low-power VLSI design techniques while integrating cryptographic hardware optimizations to address both energy efficiency and security challenges. It explores various power consumption sources in VLSI circuits and discusses techniques such as dynamic voltage scaling, clock gating, and leakage current reduction to minimize power dissipation. Additionally, it highlights cryptographic hardware power optimization techniques, including lightweight cryptography, dynamic power randomization, and secure clock gating, which are crucial for secure and energy-efficient designs.

The paper also addresses key challenges, such as the power vs. security trade-off, leakage power concerns, performance overhead, hardware complexity, reliability issues, and scalability for emerging technologies. Finally, it discusses future trends in low-power and secure VLSI, emphasizing the need for scalable, efficient, and resilient cryptographic hardware solutions in modern applications like IoT, AI, and quantum computing.

By balancing low-power design and security, this research aims to provide insights into developing next-generation VLSI architectures that meet the demands of both power efficiency and robust cryptographic security.

REFERENCES

- [1] M. Pedram, "Design technologies for low power VLSI," *Encyclopedia of Computer Science and Technology*, vol. 36, pp. 73-96, 1997.
- [2] J. Rabaey, A. Chandrakasan, and B. Nikolic, *Digital Integrated Circuits: A Design Perspective*, Prentice Hall, 2003.
- [3] P. Kocher, J. Jaffe, and B. Jun, "Differential Power Analysis," *Advances in Cryptology*, vol. 1666, pp. 388-397, 1999.
- [4] A. Bogdanov et al., "PRESENT: An Ultra-Lightweight Block Cipher," *Journal of Cryptology*, vol. 24, no. 3, pp. 446-470, 2011.
- [5] S. Roy and D. Mukhopadhyay, "Lightweight Secure Multiplier for Public Key Cryptosystems," *IEEE Transactions on Computers*, vol. 59, no. 11, pp. 1533-1545, 2010.
- [6] S. Ravi, A. Raghunathan, and N. K. Jha, "Designing secure and low-power embedded systems," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 23, no. 11, pp. 1567-1582, 2004.
- [7] J. Rajendran, G. S. Rose, and R. Karri, "Nano meets security: Exploring nanoelectronic devices for secure and energy-efficient cryptographic primitives," *Proceedings of the IEEE*, vol. 103, no. 5, pp. 829-849, 2015.
- [8] A. Chatterjee, S. Ghosh, and D. Mukhopadhyay, "A low-power physical unclonable function with enhanced challenge-response space," *IEEE Transactions on VLSI Systems*, vol. 25, no. 6, pp. 1761-1770, 2017.